

SWIFT CURRENT AP, SK, Canada

WMO: 711420

Lat: 50.290N

Lon: 107.690W

Elev: 817

StdP: 91.89

Time Zone: -6.00 (W06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-30.2	-27.1	-34.5	0.2	-30.1	-31.1	0.2	-26.9	16.1	-4.4	14.4	-7.0	5.5	270	0.778

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.8	31.2	17.6	29.2	17.1	27.1	16.5	19.6	27.1	18.5	26.0	17.5	24.8	5.5	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.2	13.6	22.4	15.9	12.5	21.2	14.9	11.7	20.2	59.8	27.0	55.8	26.1	52.5	24.6	25.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.3	11.9	10.7	DB	-33.7	34.6	2.8	1.9	-35.7	36.0	-37.3	37.1	-38.9	38.2	-40.9	39.6	(4)
(5)			WB	-33.8	22.1	2.7	1.3	-35.8	23.0	-37.4	23.8	-39.0	24.5	-41.0	25.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	4.3	-10.6	-9.5	-3.3	4.6	10.7	15.2	18.7	18.2	13.0	5.2	-2.9	-9.1	(6)
(7)		DBStd	12.28	8.84	8.01	8.08	5.70	4.81	3.60	3.27	3.88	5.03	5.53	7.21	8.15	(7)
(8)		HDD10.0	2999	640	545	411	176	51	4	0	1	26	165	387	594	(8)
(9)		HDD18.3	5259	898	778	669	413	240	106	37	51	172	407	636	852	(9)
(10)		CDD10.0	902	0	0	1	13	73	159	268	256	114	17	2	0	(10)
(11)		CDD18.3	120	0	0	0	0	3	12	47	47	12	0	0	0	(11)
(12)		CDH23.3	1584	0	0	0	6	69	156	546	619	184	6	0	0	(12)
(13)		CDH26.7	523	0	0	0	1	12	36	179	236	59	0	0	0	(13)
(14)	Wind	WSAvg	5.7	6.4	5.9	6.0	5.9	5.8	5.2	4.8	4.8	5.3	5.8	6.0	6.3	(14)
(15)	Precipitation	PrecAvg	389	22	18	21	27	47	70	48	42	36	19	16	23	(15)
(16)		PrecMax	573	50	41	71	89	133	222	109	135	99	69	45	56	(16)
(17)		PrecMin	265	3	4	3	4	7	20	7	4	3	1	3	9	(17)
(18)		PrecStd	79	12	10	17	19	31	44	28	31	27	15	11	13	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.0	7.5	15.7	23.3	28.2	30.4	33.3	34.4	31.5	24.0	16.4	8.3	(19)
(20)			MCWB	3.7	3.8	8.5	11.6	14.2	18.0	18.9	17.8	16.0	13.5	8.6	3.3	(20)
(21)		2%	DB	4.4	4.6	12.3	19.7	25.3	27.3	30.8	31.7	28.4	19.6	11.6	5.3	(21)
(22)			MCWB	2.0	1.9	6.3	10.0	13.8	16.7	18.8	17.2	14.9	11.0	6.3	2.1	(22)
(23)		5%	DB	2.9	2.8	9.2	16.8	22.6	24.9	28.7	29.4	25.0	16.3	8.6	3.4	(23)
(24)			MCWB	1.1	0.8	4.7	8.2	12.2	15.7	18.4	16.6	13.9	9.2	4.4	0.9	(24)
(25)		10%	DB	1.2	1.0	6.2	14.0	20.0	22.7	26.5	26.9	22.0	13.4	5.8	1.4	(25)
(26)			MCWB	-0.4	-0.5	2.9	7.0	11.2	14.7	17.6	16.1	12.7	7.8	2.7	-0.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.8	4.2	8.9	12.2	16.1	20.2	22.3	19.9	17.2	13.8	9.1	4.3	(27)
(28)			MCDWB	6.2	7.0	14.8	21.5	24.6	26.6	28.8	29.0	27.8	22.5	15.3	7.3	(28)
(29)		2%	WB	2.1	2.1	6.8	10.4	14.2	18.0	20.4	18.7	15.7	11.5	6.7	2.3	(29)
(30)			MCDWB	4.1	4.1	11.6	18.3	23.4	24.3	27.3	27.1	25.2	17.9	11.2	5.0	(30)
(31)		5%	WB	1.1	0.9	4.9	8.9	13.1	16.6	19.1	17.8	14.5	9.7	4.6	0.9	(31)
(32)			MCDWB	2.9	2.6	9.0	15.9	21.0	22.8	26.3	26.2	23.5	15.4	8.2	3.3	(32)
(33)		10%	WB	-0.4	-0.5	3.1	7.4	11.8	15.5	18.1	16.9	13.2	8.1	2.8	-0.5	(33)
(34)			MCDWB	1.2	0.9	6.1	13.2	18.6	21.2	25.1	24.9	21.1	12.8	5.6	1.4	(34)
(35)	Mean Daily Temperature Range	MDBR		8.9	9.0	9.5	11.7	12.5	11.1	12.8	13.4	12.5	10.8	8.9	8.4	(35)
(36)		5% DB	MCDBR	9.5	9.7	13.0	16.1	16.3	14.8	16.0	17.5	17.4	15.3	12.2	9.6	(36)
(37)			MCWBR	7.8	7.7	8.1	8.5	7.8	7.1	7.1	6.8	7.7	8.4	8.0	7.3	(37)
(38)		5% WB	MCDWB	9.4	9.6	12.4	15.0	14.7	13.0	14.0	14.5	15.5	14.1	11.9	9.7	(38)
(39)			MCWBR	7.9	7.7	8.0	8.4	7.3	6.8	7.1	6.3	7.2	8.1	8.0	7.6	(39)
(40)	Clear-Sky Solar Irradiance	taub		0.245	0.246	0.276	0.343	0.364	0.357	0.364	0.383	0.333	0.286	0.256	0.234	(40)
(41)		taud		2.417	2.484	2.503	2.322	2.309	2.386	2.381	2.291	2.407	2.530	2.500	2.445	(41)
(42)		Ebn at Noon		826	925	946	900	892	899	886	848	862	858	808	793	(42)
(43)		Edh at Noon		59	72	85	115	122	114	113	118	94	69	54	49	(43)
(44)	All-Sky Solar Radiation	RadAvg		1.21	2.22	3.55	4.84	5.82	6.15	6.68	5.38	3.94	2.41	1.32	0.93	(44)
(45)		RadStd		0.13	0.17	0.26	0.35	0.41	0.42	0.41	0.41	0.36	0.28	0.15	0.09	(45)

Historical Trends

		DBAvg		Heating		Cooling		Degree-Days			
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (16 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page